Minqi's Solutions to the Exercises of Sheldon Axler (2014), Linear Algebra Done Right (3rd Edition)

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CHAPTER 1

Vector Spaces

1.1. Exercise 1.A.1

Suppose a and b are real numbers, not both 0. Find real numbers c and d such that

$$1/(a+bi) = c + di$$

Answer. Since at least one of the real numbers a, b is different from 0, we have $a^2 + b^2 \neq 0$. We can then let

$$c = \frac{a}{a^2+b^2},$$

$$d = \frac{-b}{a^2+b^2}.$$

Then

$$\begin{aligned}
&(a+bi)\left(\frac{a}{a^2+b^2} - \frac{b}{a^2+b^2}i\right) \\
&= \left(a\frac{a}{a^2+b^2} - b\left(-\frac{b}{a^2+b^2}\right)\right) + \left(a\left(-\frac{b}{a^2+b^2}\right) + b\frac{a}{a^2+b^2}\right)i \\
&= 1 + 0i \\
&= 1.
\end{aligned}$$

□

1.2. Exercise 1.A.2

Show that

$$\frac{-1+\sqrt{3}i}{2}$$

is a cube root of 1 (meaning that its cube equals 1).

Answer.

$$\begin{aligned}
&\frac{-1+\sqrt{3}i}{2} * \frac{-1+\sqrt{3}i}{2} * \frac{-1+\sqrt{3}i}{2} \\
&= \frac{-2-2\sqrt{3}i}{4} * \frac{-1+\sqrt{3}i}{2} \\
&= \frac{(-2)(-1) - (-2\sqrt{3})(\sqrt{3}) + (-2\sqrt{3} + (-2\sqrt{3})(-1))i}{8} \\
&= 1
\end{aligned}$$

□

1.3. Exercise 1.A.3

Find two distinct square roots of i.

Answer. Note that

$$i = e^{i\frac{\pi}{2}}.$$

Thus the following two numbers are two distinct square roots of i:

$$\pm e^{i\frac{\pi}{4}} = \pm(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i).$$

□

1.4. Exercise 1.A.4

Show that $\alpha + \beta = \beta + \alpha$ for all $\alpha, \beta \in \mathbb{C}$.

Proof. Suppose $\alpha = a + bi$ and $\beta = c + di$, where $a, b, c, d \in \mathbb{R}$. Then the definition of addition of complex numbers shows that

$$\alpha + \beta = (a + c) + (b + d)i$$

and

$$\beta + \alpha = (c + a) + (d + b)i.$$

The equations above and the commutativity of addition of real numbers show that $\alpha + \beta = \beta + \alpha$. □

1.5. Exercise 1.A.5

Show that $(\alpha + \beta) + \lambda = \alpha + (\beta + \lambda)$ for all $\alpha, \beta, \lambda \in \mathbb{C}$.

Proof. Suppose $\alpha = a+bi$, $\beta = c+di$ and $\lambda = e+fi$, where $a, b, c, d, e, f \in \mathbb{R}$. Then the definition of addition of complex numbers shows that

$$\begin{aligned}(\alpha + \beta) + \lambda &= ((a + c) + (b + d)i) + e + fi \\ &= ((a + c) + e) + ((b + d) + f)i\end{aligned}$$

and

$$\begin{aligned}\alpha + (\beta + \lambda) &= a + bi + ((c + e) + (d + f)i) \\ &= (a + (c + e)) + (b + (d + f))i.\end{aligned}$$

The equations above and the associativity of addition of real numbers show that $(\alpha + \beta) + \lambda = \alpha + (\beta + \lambda)$. □

1.6. Exercise 1.A.6

Show that $(\alpha\beta)\lambda = \alpha(\beta\lambda)$ for all $\alpha, \beta, \lambda \in \mathbb{C}$.

Proof. Suppose $\alpha = a+bi$, $\beta = c+di$ and $\lambda = e+fi$, where $a, b, c, d, e, f \in \mathbb{R}$. Then the definition of multiplication of complex numbers shows that

$$\begin{aligned}(\alpha\beta)\lambda &= ((ac - bd) + (ad + bc)i)(e + fi) \\ &= ((ac - bd)e - (ad + bc)f) + ((ac - bd)f + (ad + bc)e)i \\ &= (ace - bde - adf - bcf) + (acf - bdf + ade + bce)i\end{aligned}$$

and

$$\begin{aligned}\alpha(\beta\lambda) &= (a+bi)((ce-df)+(cf+de)i)\\ &= (a(ce-df)-b(cf+de))+(a(cf+de)+b(ce-df))i\\ &= (ace-adf-bcf-bde)+(acf+ade+bce-bdf)i.\end{aligned}$$

The equations above and the commutativity of addition of real numbers show that $(\alpha\beta)\lambda = \alpha(\beta\lambda)$. □

1.7. Exercise 1.A.7: Existence and Uniqueness of Additive Inverse in $\mathbb{C}$

Show that for every $\alpha \in \mathbb{C}$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha + \beta = 0$.

Proof. Suppose $\alpha = a+bi$ where $a, b \in \mathbb{R}$, then $\beta = -a-bi$ satisfies $\alpha+\beta = 0$. Suppose that there exists another $\beta' \in \mathbb{C}$ such that $\alpha + \beta' = 0$. Then

$$\begin{aligned}\beta &= \beta + (\alpha + \beta')\\ &= (\beta + \alpha) + \beta'\\ &= \beta'.\end{aligned}$$

The equations above show that $\beta' = \beta$. □

1.8. Exercise 1.A.8: Existence and Uniqueness of Multiplicative Inverse in $\mathbb{C}$

Show that for every $\alpha \in \mathbb{C}$ with $\alpha \neq 0$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha\beta = 1$.

Proof. The existence of β has been shown in Exercise 1.A.1.
Suppose that there exists another $\beta' \in \mathbb{C}$ such that and $\alpha\beta' = 1$. Then

$$\begin{aligned}\beta &= \beta(\alpha\beta')\\ &= (\beta\alpha)\beta'\\ &= \beta'.\end{aligned}$$

The equations above show that $\beta' = \beta$. □

1.9. Exercise 1.A.9

Show that $\lambda(\alpha + \beta) = \lambda\alpha + \lambda\beta$ for all $\lambda, \alpha, \beta \in \mathbb{C}$.

Proof. Suppose $\alpha = a+bi$, $\beta = c+di$ and $\lambda = e+fi$, where $a, b, c, d, e, f \in \mathbb{R}$. Then the definition of multiplication and addition of complex numbers shows that

$$\begin{aligned}\lambda(\alpha+\beta) &= (e+fi)((a+c)+(b+d)i)\\ &= (e(a+c)-f(b+d))+(e(b+d)+f(a+c))i\\ &= (ea+ec-fb-fd)+(eb+ed+fa+fc)i\end{aligned}$$

and

$$\begin{aligned}\lambda\alpha+\lambda\beta &= (e+fi)(a+bi)+(e+fi)(c+di)\\ &= ((ea-fb)+(eb+fa)i)+((ec-fd)+(ed+fc)i)\\ &= (ea-fb+ec-fd)+(eb+fa+ed+fc)i.\end{aligned}$$

The equations above and the commutativity of addition of real numbers show that $\lambda(\alpha+\beta) = \lambda\alpha+\lambda\beta$. □

1.10. Exercise 1.A.10: A Vector Addition and Scalar Multiplication Example in $\mathbb{R}^4$

Find $x \in \mathbb{R}^4$ such that

$$(4, -3, 1, 7) + 2x = (5, 9, -6, 8).$$

Answer.

$$x = \left(\frac{1}{2}, 6, -\frac{7}{2}, \frac{1}{2}\right).$$

□

1.11. Exercise 1.A.11: A Scalar Multiplication Example in $\mathbb{C}^3$

Explain why there does not exist $\lambda \in \mathbb{C}$ such that

$$\lambda(2 - 3i, 5 + 4i, -6 + 7i) = (12 - 5i, 7 + 22i, -32 - 9i).$$

Proof. If there exists such λ then the definition of scalar multiplication in $\mathbb{C}^3$ shows that,

$$\lambda(2 - 3i) = 12 - 5i.$$

Thus

$$\lambda = \frac{12 - 5i}{2 - 3i} = 3 + 2i.$$

But

$$\begin{aligned}\lambda(2 - 3i, 5 + 4i, -6 + 7i) &= (3 + 2i)(2 - 3i, 5 + 4i, -6 + 7i)\\ &= (12 - 5i, 7 + 22i, -32 + 9i)\\ &\neq (12 - 5i, 7 + 22i, -32 - 9i).\end{aligned}$$

Therefore λ does not exist. □

1.12. Exercise 1.A.12: Associativity of Vector Addition in $\mathbb{F}^n$

Show that $(x + y) + z = x + (y + z)$ for all $x, y, z \in \mathbb{F}^n$.

Proof. Suppose $x = (x_1, \ldots, x_n)$, $y = (y_1, \ldots, y_n)$ and $z = (z_1, \ldots, z_n)$ where $x_1, \ldots, x_n, y_1, \ldots, y_n, z_1, \ldots, z_n \in \mathbb{F}$. Then the definition of addition in $\mathbb{F}^n$ shows that

$$\begin{aligned}(x + y) + z &= (x_1 + y_1, \ldots, x_n + y_n) + (z_1, \ldots, z_n)\\ &= ((x_1 + y_1) + z_1, \ldots, (x_n + y_n) + z_n)\end{aligned}$$

and

$$\begin{aligned}x + (y + z) &= (x_1, \ldots, x_n) + (y_1 + z_1, \ldots, y_n + z_n)\\ &= (x_1 + (y_1 + z_1), \ldots, x_n + (y_n + z_n))\end{aligned}$$

The equations above and the associativity of addition in $\mathbb{F}$ show that $(x + y) + z = x + (y + z)$. □

1.13. Exercise 1.A.13: Associativity of Scalar Multiplication in $\mathbb{F}^n$

Show that $(ab)x = a(bx)$ for all $x \in \mathbb{F}^n$ and all $a, b \in \mathbb{F}$.

PROOF. Suppose $x = (x_1, \ldots, x_n)$ where $x_1, \ldots, x_n \in \mathbb{F}$. Then the definition of scalar multiplication in $\mathbb{F}^n$ shows that

$$(ab)x = ((ab)x_1, \ldots, (ab)x_n)$$

and

$$\begin{aligned} a(bx) &= a(bx_1, \ldots, bx_n) \\ &= (a(bx_1), \ldots, a(x_n)) \end{aligned}$$

The equations above and the associativity of multiplication in $\mathbb{F}$ show that $(ab)x = a(bx)$. □

1.14. Exercise 1.A.14: Scalar Multiplication by 1 in $\mathbb{F}^n$

Show that $1x = x$ for all $x \in \mathbb{F}^n$.

PROOF. Suppose $x = (x_1, \ldots, x_n)$ where $x_1, \ldots, x_n \in \mathbb{F}$. Then the definition of scalar multiplication in $\mathbb{F}^n$ and the fact that 1 is the multiplicative identity of $\mathbb{F}$ shows that

$$\begin{aligned} 1x &= (1x_1, \ldots, 1x_n) \\ &= (x_1, \ldots, x_n) \\ &= x. \end{aligned}$$

□

1.15. Exercise 1.A.15: Distributing Vector Additions with Scalar Multiplications in $\mathbb{F}^n$

Show that $\lambda(x + y) = \lambda x + \lambda y$ for all $\lambda \in \mathbb{F}$ and $x, y \in \mathbb{F}^n$.

PROOF. Suppose $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$ where $x_1, \ldots, x_n, y_1, \ldots, y_n \in \mathbb{F}$. Then the definition of addition and scalar multiplication in $\mathbb{F}^n$ shows that

$$\begin{aligned} \lambda(x + y) &= \lambda(x_1 + y_1, \ldots, x_n + y_n) \\ &= (\lambda(x_1 + y_1), \ldots, \lambda(x_n + y_n)) \end{aligned}$$

and

$$\begin{aligned} \lambda x + \lambda y &= (\lambda x_1, \ldots, \lambda x_n) + (\lambda y_1, \ldots, \lambda y_n) \\ &= (\lambda x_1 + \lambda y_1, \ldots, \lambda x_n + \lambda y_n) \end{aligned}$$

The equations above and the distributive property of $\mathbb{F}$ show that $\lambda(x + y) = \lambda x + \lambda y$. □

1.16. Exercise 1.A.16: Distributing Scalar Additions with Scalar Multiplications in $\mathbb{F}^n$

Show that $(a+b)x = ax + bx$ for all $a, b \in \mathbb{F}$ and all $x \in \mathbb{F}^n$.

PROOF. Suppose $x = (x_1, \ldots, x_n)$ where $x_1, \ldots, x_n \in \mathbb{F}$. Then the definition of scalar multiplication and addition in $\mathbb{F}^n$ shows that

$$(a+b)x = ((a+b)x_1, \ldots, (a+b)x_n)$$

and

$$\begin{aligned} ax + bx &= (ax_1, \ldots, ax_n) + (bx_1, \ldots, bx_n) \\ &= (ax_1 + bx_1, \ldots, ax_n + bx_n). \end{aligned}$$

The equations above and the distributive property of $\mathbb{F}$ show that $(a+b)x = ax + bx$. □

1.17. Exercise 1.B.1

Prove that $-(-v) = v$ for every $v \in V$.

PROOF.

$$\begin{aligned} -(-v) &= -(-v) + 0 \\ &= -(-v) + ((-v) + v) \\ &= (-(-v) + (-v)) + v \\ &= 0 + v \\ &= v \end{aligned}$$

□

1.18. Exercise 1.B.2

Suppose $a \in \mathbb{F}, v \in V$, and $av = 0$. Prove that $a = 0$ or $v = 0$.

PROOF. For the purpose of arriving at a contradiction, assume that both $a \neq 0$ and $v \neq 0$. Then

$$v = 1v = (\frac{1}{a}a)v = \frac{1}{a}(av) = 0$$

which contradicts with $v \neq 0$.

Therefore $a = 0$ or $v = 0$. □

1.19. Exercise 1.B.3

Suppose $v, w \in V$. Explain why there exists a unique $x \in V$ such that $v + 3x = w$.

PROOF. We claim that the following $x \in V$ satisfies $v + 3x = w$:

$$x = \frac{1}{3}(w - v).$$

We can verify it by showing that,

$$\begin{aligned} v + 3x &= v + 3 \times \frac{1}{3}(w - v) \\ &= v + w - v \\ &= w. \end{aligned}$$

Suppose that there exists another $x' \in V$ that also satisfies $v + 3x' = w$, then

$$\begin{aligned} 0 &= w - w \\ &= (v + 3x) - (v + 3x') \\ &= v + 3x - v - 3x' \\ &= 3x - 3x'. \end{aligned}$$

Thus

$$\begin{aligned} x' &= x' + \frac{1}{3} \times 0 \\ &= x' + \frac{1}{3}(3x - 3x') \\ &= x' + x - x' \\ &= x. \end{aligned}$$

□

1.20. Exercise 1.B.4

The empty set is not a vector space. The empty set fails to satisfy only one of the requirements listed in 1.19. Which one?

Answer. The empty set fails to satisfy the "Additive Identity" requirement. Because there does not exist an element $0 \in \varnothing$. □

1.21. Exercise 1.B.5

Show that in the definition of a vector space (1.19), the additive inverse condition can be replaced with the condition that

$$0v = 0, \forall v \in V.$$

Here the 0 on the left side is the number 0, and the 0 on the right side is the additive identity of V.

Proof. We will show that the new condition is equal to the original condition, which states that for every $v \in V$, there exists $w \in V$ such that $v + w = 0$. That the old condition implies the new condition has been shown in Theorem 1.29 "The number 0 times a vector". Therefore it suffices to show that the new condition implies the old condition.

Suppose that the new condition holds.

Pick $v \in V$, and let $w = (-1)v$. It follows that

$$\begin{aligned} v + w &= 1v + (-1)v \\ &= (1 - 1)v \\ &= 0v \\ &= 0. \end{aligned}$$

Therefore the old condition holds. □

1.22. Exercise 1.B.6

Let ∞ and $-\infty$ denote two distinct objects, neither of which is in $\mathbb{R}$. Define an addition and scalar multiplication on $\mathbb{R} \cup \{\infty\} \cup \{-\infty\}$ as you could guess from the notation. Specifically, the sum and product of two real numbers is as usual, and for $t \in \mathbb{R}$ define

$$t\infty = \begin{cases} -\infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ \infty & \text{if } t > 0, \end{cases} \quad t(-\infty) = \begin{cases} \infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ -\infty & \text{if } t > 0, \end{cases}$$

$$t + \infty = \infty + t = \infty, \quad t + (-\infty) = (-\infty) + t = (-\infty),$$

$$\infty + \infty = \infty, \quad (-\infty) + (-\infty) = (-\infty), \quad \infty + (-\infty) = 0.$$

Is $\mathbb{R} \cup \{\infty\} \cup \{-\infty\}$ a vector space over $\mathbb{R}$? Explain.

Answer. No. Because the associativity of vector addition fails:

$$(1024 + \infty) - \infty = \infty - \infty = 0$$

but

$$1024 + (\infty - \infty) = 1024 + 0 = 1024.$$

□

1.23. Exercise 1.C.1

For each of the following subsets of $\mathbb{F}^3$, determine whether it is a subspace of $\mathbb{F}^3$:

(a) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$;
(b) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 4\}$;
(c) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 x_2 x_3 = 0\}$;
(d) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}$.

Answer. (a) Denote $A = \{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$. We claim that A is a subspace of $\mathbb{F}^3$ with the following verifications.

(i) $(0, 0, 0) \in A$ because $0 + 2 * 0 + 3 * 0 = 0$.

(ii) Pick $u, w \in A$, where $u = (u_1, u_2, u_3), w = (w_1, w_2, w_3)$. It follows that $u + w = (u_1 + w_1, u_2 + w_2, u_3 + w_3)$. We can show that $u + w \in A$ by the following equations:

$$\begin{aligned} &(u_1 + w_1) + 2(u_2 + w_2) + 3(u_3 + w_3) \\ &= (u_1 + 2u_2 + 3u_3) + (w_1 + 2w_2 + 3w_3) \\ &= 0 + 0 = 0. \end{aligned}$$

(iii) Pick $a \in \mathbb{F}$ and $u \in A$, where $u = (u_1, u_2, u_3)$. It follows that $au = (au_1, au_2, au_3)$. We can show that $au \in A$ by the following equations:

$$\begin{aligned} (au_1) + 2(au_2) + 3(au_3) &= a(u_1 + 2u_2 + 3u_3) \\ &= a0 = 0. \end{aligned}$$

(b) Denote $B = \{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 4\}$. B is not a subspace of $\mathbb{F}^3$, because vector $(0, 0, 0) \notin B$ (as $0 + 2 * 0 + 3 * 0 \neq 4$).

(c) Denote $C = \{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1x_2x_3 = 0\}$. C is not a subspace of $\mathbb{F}^3$ because by setting $u = (0, 1, 1) \in C$ and $w = (1, 1, 0) \in C$, we find out that $u + w = (1, 2, 1) \notin C$.

(d) Denote $D = \{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}$. We claim that D is a subspace of $\mathbb{F}^3$ with the following verifications.

(i) $(0, 0, 0) \in A$ because $0 = 5 * 0$.

(ii) Pick $u, w \in A$, where $u = (u_1, u_2, u_3), w = (w_1, w_2, w_3)$. It follows that $u+w = (u_1+w_1, u_2+w_2, u_3+w_3)$. We can show that $u+w \in A$ by the following equations:

$$\begin{aligned} u_1 + w_1 &= 5u_3 + 5w_3 \\ &= 5(u_1 + w_3) \end{aligned}$$

(iii) Pick $a \in \mathbb{F}$ and $u \in A$, where $u = (u_1, u_2, u_3)$. It follows that $au = (au_1, au_2, au_3)$. We can show that $au \in A$ by the following equations:

$$\begin{aligned} au_1 &= a(5u_3) \\ &= 5(au_3). \end{aligned}$$

□

1.24. Exercise 1.C.2

Verify all the assertions in Example 1.35.

(a) If $b \in \mathbb{F}$, then

$$\{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 : x_3 = 5x_4 + b\}$$

is a subspace of $\mathbb{F}^4$ if and only if $b = 0$.

(b) The set of continuous real-values functions on the interval $[0, 1]$ is a subspace of $\mathbb{R}^{[0,1]}$.

(c) The set of differentiable real-valued functions on $\mathbb{R}$ is a subspace of $\mathbb{R}^{\mathbb{R}}$

(d) The set of differentiable real-valued functions f on the interval $(0, 3)$ such that $f'(2) = b$ is a subspace of $\mathbb{R}^{(0,3)}$ if and only if $b = 0$.

(e) The set of all sequences of complex numbers with limit 0 is a subspace of $\mathbb{C}^\infty$.

Answer. (a) Denote $A = \{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 : x_3 = 5x_4 + b\}$.

If $b = 0$, then A is a subspace by a verification similar to Exercise 1.C.2 (d).

On the other hand, if A is a subspace, then $0 \in A$. It follows that

$$0 = 5 * 0 + b = b.$$

(b) Denote $B = \{f \in \mathbb{R}^{[0,1]} : f \text{ is continuous}\}$.

We can verify that $0 \in B$ because the constant function $f = 0$ is continuous.

Suppose that $u, w \in B$, we can verify that $(u+w) \in B$ because the addition of two complex continuous functions is continuous by Theorem 4.9 of [**?**].

Suppose that $a \in \mathbb{F}$ and $u \in B$. Note that the constant function $f = a$ is continuous. We can verify that $au \in B$ because the multiplication of two continuous complex functions is continuous by Theorem 4.9 of [**?**].

(c) Denote $C = \{f \in \mathbb{R}^{\mathbb{R}} : f \text{ is differentiable}\}$.

We can verify that $0 \in C$ because the constant function $f = 0$ is differentiable.

Suppose that $u, w \in C$, we can verify that $(u+w) \in C$ because the addition of two differentiable real functions is differentiable by Theorem 5.3 of [**?**].

Suppose that $a \in \mathbb{R}$ and $u \in C$. Note that the constant function $a(x) = a$ is differentiable. We can verify that $au \in C$ because the multiplication of two differentiable real functions is differentiable by Theorem 5.3 of [**?**].

(d) Denote $D = \{f \in \mathbb{R}^{(0,3)} : f \text{ is differentiable and } f'(2) = b\}$.

Suppose that $b = 0$. We claim that D is a subspace of $\mathbb{R}^{(0,3)}$ with the following verifications.

(i) $0 \in D$ because the constant function $0(x) = 0$ is differentiable and $0'(2) = 0$.

(ii) Suppose that $u, w \in D$. $(u+w)$ is also differentiable by Theorem 5.3 of [**?**]. Also by Theorem 5.3 of [**?**], $(u+v)'(2) = u'(2) + v'(2) = 0 + 0 = 0$. Thus $(u+w) \in D$.

(iii) Suppose that $a \in \mathbb{R}$ and $u \in D$. Note that the constant function $a(x) = a$ is differentiable, so au is differentiable by Theorem 5.3 of [**?**]. Also by Theorem 5.3 of [**?**], $(au)'(2) = a'(2)u(2) + a(2)u'(2) = 0 \times u(2) + a \times 0 = 0$. Thus $au \in D$.

On the other hand, suppose that D is a subspace. Then it implies that $0 \in D$ where $0(x) = 0$, and $0'(x) = 0$ implies that $0'(2) = b = 0$.

(e) Denote $E = \{z_n \in \mathbb{C}^{\infty} : \lim_{n\to\infty} z_n = 0\}$.

We can verify that $0_n \in E$ where $0_n = (0, 0, 0, \dots)$ because $\lim_{n\to\infty} 0_n = 0$.

Suppose that $u_n, w_n \in E$, we can verify that $(u_n + w_n) \in C$ because $\lim_{n\to\infty}(u_n + w_n) = \lim_{n\to\infty} u_n + \lim_{n\to\infty} w_n = 0 + 0 = 0$ by Theorem 3.3 of [**?**].

Suppose that $a \in \mathbb{F}$ and $u_n \in E$, we can verify that $au_n \in E$ because $\lim_{n\to\infty} au_n = a \lim_{n\to\infty} u_n = a0 = 0$ by Theorem 3.3 of [**?**].

□

1.25. Exercise 1.C.3

Show that the set of differentiable real-valued functions f on the interval $(-4, 4)$ such that $f'(-1) = 3f(2)$ is a subspace of $\mathbb{R}^{(-4,4)}$.

Proof. Denote $X = \{f \in \mathbb{R}^{(-4,4)} : f \text{ is differentiable and } f'(-1) = 3f(2)\}$

(1) We claim that $0 \in X$ where $0(x) = 0$. Note that 0 is differentiable and $0'(-1) = 0, 0(2) = 0$, thus $0'(-1) = 3 \times 0(2)$.

(2) Pick $f, g \in X$. That $f + g$ is differentiable follows from Theorem 5.3 of [**?**]. Also by Theorem 5.3 of [**?**], $(f+g)'(-1) = f'(-1) + g'(-1) = 3f(2) + 3g(2) = 3(f(2) + g(2)) = 3((f+g)(2))$. Thus $f + g \in X$.

(3) Pick $f \in X$ and $a \in \mathbb{F}$. That af is differentiable follows from Theorem 5.3 of [**?**] by considering a as a constant function $a(x) = a$. Also by Theorem 5.3 of [**?**], $(af)'(-1) = a(f'(-1)) = a(3f(2)) = 3(af(2)) = 3((af)(2))$. Thus $af \in X$.

□

1.26. Exercise 1.C.4

Suppose $b \in \mathbb{R}$. Show that the set of continuous real-valued functions f on the interval $[0,1]$ such that $\int_0^1 f = b$ is a subspace of $\mathbb{R}^{[0,1]}$ if and only if $b = 0$.

Proof. Denote $X = \{f \in \mathbb{R}^{[0,1]} :$f is continuous and $\int_0^1 f = b\}$.
Suppose that $b = 0$. We now verify that X is a subspace.

(1) We can verify that $0 \in X$ because $0(x) = 0$ is continuous and $\int_0^1 0(x)dx = 0$.
(2) Pick $f, g \in X$. It follows that $f + g$ is continuous by Theorem 4.9 of [**?**] and Riemann-integrable by Theorem 6.8 of [**?**]. Further by Theorem 6.12 of [**?**], we have $\int_0^1 (f(x) + g(x))dx = \int_0^1 f(x)dx + \int_0^1 g(x) = 0 + 0 = 0$.
(3) Pick $a \in \mathbb{F}, f \in X$. By Theorem 4.9 of [**?**], af is continuous. Thus af is Riemann-integrable by Theorem 6.8 of [**?**]. Further by Theorem 6.12 of [**?**], we have $\int_0^1 (af(x))dx = a\int_0^1 f(x)dx = a \times 0 = 0$.

On the other hand, suppose that X is a subspace. Then it follows that $0 \in X$, which implies that $\int_0^1 0(x)dx = 0 = b$. □

1.27. Exercise 1.C.5

Is $\mathbb{R}^2$ a subspace of the complex vector space $\mathbb{C}^2$?

Answer. No. Because $\mathbb{R}^2 \not\subset \mathbb{C}^2$. □

1.28. Exercise 1.C.6

(i) Is $\{(a, b, c) \in \mathbb{R}^3 : a^3 = b^3\}$ a subspace of $\mathbb{R}^3$?
(ii) Is $\{(a, b, c) \in \mathbb{C}^3 : a^3 = b^3\}$ a subspace of $\mathbb{C}^3$?

Answer. (i) Yes. Let's observe the real function $f(x) = x^3$. $f(x)$ is a 1-1 correspondence between $\mathbb{R}$ and $\mathbb{R}$. Therefore the condition $a^3 = b^3$ is equivalent to $a = b$ in $\mathbb{R}$.
(ii) No. Set $u = (2, 2e^{\frac{2\pi}{3}i}, 0)$ and $v = (e^{\frac{2\pi}{3}i}, 1, 0)$. Then $u + v = (2 + e^{\frac{2\pi}{3}i}, 2e^{\frac{2\pi}{3}i} + 1, 0)$ where

$$\left(2 + e^{\frac{2\pi}{3}i}\right)^3 = 3\sqrt{3}i,$$

$$\left(2e^{\frac{2\pi}{3}i} + 1\right)^3 = -3\sqrt{3}i.$$

□

1.29. Exercise 1.C.7

Give an example of a nonempty subset U of $\mathbb{R}^2$ such that U is closed under addition and under taking additive inverses (meaning $-u \in U$ whenever $u \in U$), but U is not a subspace of $\mathbb{R}^2$.

Answer.

$$U = \{(0, 0), (1, 1), (-1, -1)\}.$$

□

1.30. Exercise 1.C.8

Give an example of a nonempty subset U of $\mathbb{R}^2$ such that U is closed under scalar multiplication, but U is not a subspace of $\mathbb{R}^2$.

Answer.
$$U = \{(0, y) : y \in \mathbb{R}\} \cup \{(x, 0) : x \in \mathbb{R}\}.$$
□

1.31. Exercise 1.C.9

A function $f : \mathbb{R} \to \mathbb{R}$ is called periodic if there exists a positive number p such that $f(x) = f(x + p)$ for all $x \in \mathbb{R}$. Is the set of periodic functions from $\mathbb{R}$ to $\mathbb{R}$ a subspace of $\mathbb{R}^{\mathbb{R}}$? Explain.

Lemma 1.1. *$f(x) = \cos(\pi x) + \cos(x)$ is not periodic.*

Proof. Suppose that $f(x) = \cos(\pi x) + \cos(x)$ is periodic, then there exists $\theta \in \mathbb{R}$ such that $\theta > 0$ and $f(x + \theta) = f(x)$ for all $x \in \mathbb{R}$. Pick $x = 0$, we have

$$f(\theta + 0) = \cos(\pi\theta) + \cos(\theta) = f(0) = \cos(0) + \cos(0) = 2. \tag{1.1}$$

Note that $-1 \leqslant \cos(\pi\theta) \leqslant 1$ and $-1 \leqslant \cos(\theta) \leqslant 1$. However, if $\cos(\pi\theta) < 1$ and $\cos(\theta) < 1$, then $\cos(\pi\theta) + \cos(\theta) < 1$. Thus Equation (1.1) implies that
$$\begin{cases} \cos(\pi\theta) = 1 \\ \cos(\theta) = 1 \end{cases}.$$
This further implies that there exists some $m, n \in \mathbb{Z}$ such that,
$$\begin{cases} \pi\theta = 2m\pi \\ \theta = 2n\pi \end{cases}.$$
Since $\theta \neq 0$, dividing the above two equations yields
$$\pi = \frac{m}{n}.$$
This contradicts with the fact that π is irrational. □

Main Answer. No, because Lemma 1.1 gives an example of two periodic functions where their addition is not periodic. □

1.32. Exercise 1.C.10

Suppose U_1 and U_2 are subspaces of V. Prove that the intersection $U_1 \cap U_2$ is a subspace of V.

Proof. We prove by verifying the three conditions for $U_1 \cap U_2$ being a subspace.

(1) The premise that U_1 is a subspace implies $0 \in U_1$. The premise that U_2 is a subspace implies $0 \in U_2$. Therefore $0 \in U_1 \cap U_2$.
(2) Pick $u, w \in U_1 \cap U_2$. The premise that U_1 is a subspace implies $u + w \in U_1$. The premise that U_2 is a subspace implies $u + w \in U_2$. Therefore $u + w \in U_1 \cap U_2$.
(3) Pick $a \in \mathbb{F}$ and $u \in U_1 \cap U_2$. The premise that U_1 is a subspace implies that $au \in U_1$. The premise that U_2 is a subspace implies $au \in U_2$. Therefore $au \in U_1 \cap U_2$.

□

1.33. Exercise 1.C.11

Prove that the intersection of every collection of subspaces of V is a subspace of V.

Proof. Pick set I and $(S_i)_{i\in I}$, a family of sets indexed over set I. Suppose that S_i is a subspace of V for all $i \in I$. We prove by verifying the three conditions for $\cap_{i\in I} S_i$ being a subspace.

(1) For all $i \in I$, the premise that S_i is a subspace implies that $0 \in S_i$. Therefore $0 \in \cap_{i\in I} S_i$.
(2) Pick $u, w \in \cap_{i\in I} S_i$. For all $i \in I$, the premise that S_i is a subspace implies $u + w \in S_i$. Therefore $u + w \in \cap_{i\in I} S_i$.
(3) Pick $a \in \mathbb{F}$ and $u \in \cap_{i\in I} S_i$. For all $i \in I$, the premise that U_i is a subspace implies that $au \in U_i$. Therefore $au \in \cap_{i\in I} S_i$.

□

1.34. Exercise 1.C.12

Prove that the union of two subspaces of V is a subspace of V if and only if one of the subspaces is contained in the other.

Proof. Pick two subspaces V_1, V_2 of V.

(1) If $V_1 \subset V_2$, then $V_1 \cup V_2 = V_2$. Since V_2 is a subspace, it follows that $V_1 \cup V_2$ is a subspace.
(2) Suppose that $V_1 \cup V_2$ is a subspace. To prove by contradiction, suppose that $V_1 \not\subset V_2$ and $V_2 \not\subset V_1$. Thus there exists $v_1 \in V_1$ such that $v_1 \notin V_2$, and there exists $v_2 \in V_2$ such that $v_2 \notin V_1$.

    We claim that $v_1 + v_2 \notin V_1$. Because if $v_1 + v_2$ was in V_1, the premise that V_1 is a subspace would imply that $v_2 = (v_1 + v_2) - v_1$ is also in V_1.

    We claim that $v_1 + v_2 \notin V_2$. Because if $v_1 + v_2$ was in V_2, the premise that V_2 is a subspace would imply that $v_1 = (v_1 + v_2) - v_2$ is also in V_2.

    Thus $v_1 + v_2 \notin V_1 \cup V_2$, contradicting with the premise that $V_1 \cup V_2$ is a subspace.

□

1.35. Exercise 1.C.13

Prove that the union of three subspaces of V is a subspace of V if and only if one of the subspaces contains the other two.

Proof. Pick three subspaces V_1, V_2, V_3 of V.

(1) If $V_1 \subset V_3$ and $V_2 \subset V_3$, then $V_1 \cup V_2 \cup V_3 = V_3$. Since V_3 is a subspace, it follows that $V_1 \cup V_2 \cup V_3$ is a subspace.
(2) Suppose that $V_1 \cup V_2 \cup V_3$ is a subspace of V.

    If $V_1 \subset V_2$, then $V_1 \cup V_2 \cup V_3$ becomes $V_2 \cup V_3$. By Exercise 1.C.12, it follows that either $V_2 \subset V_3$ or $V_3 \subset V_2$. Thus either V_3 or V_2 is the subspace that contains the other two.

    If $V_2 \subset V_1$, then $V_1 \cup V_2 \cup V_3$ becomes $V_1 \cup V_3$. By Exercise 1.C.12, it follows that either $V_1 \subset V_3$ or $V_3 \subset V_1$. Thus either V_3 or V_1 is the subspace that contains the other two.

If $V_1 \not\subset V_2$ and $V_2 \not\subset V_1$, then there exists $u \in V_1$ and $w \in V_2$ such that $u \notin V_2$ and $w \notin V_1$.

(i) We claim that $V_1 \setminus (V_1 \cap V_2) \subset V_3$. Pick $x \in V_1 \setminus (V_1 \cap V_2) \subset V_3$. The premise that $V_1 \cup V_2 \cup V_3$ is a subspace implies that $x+w \in V_1 \cup V_2 \cup V_3$ and $2x + w \in V_1 \cup V_2 \cup V_3$. If $x + w$ was in V_1, then $w = (x + w) - x$ would be in V_1 which is a contradiction. If $2x + w$ was in V_1, then $w = (2x+w)-2x$ would be in V_1 which is a contradiction. If $x+w$ was in V_2, then $x = (x + w) - w$ would be in V_2 which is a contradiction. If $2x + w$ was in V_2, then $2x = (2x + w) - w$ would be in V_2 which is a contradiction. Therefore $x + w \in V_3$ and $2x + w \in V_3$. And it follows that $x = (2x + w) - (x + w)$ is in V_3.

(ii) We claim that $V_2 \setminus (V_1 \cap V_2) \subset V_3$. Pick $y \in V_2 \setminus (V_1 \cap V_2) \subset V_3$. The premise that $V_1 \cup V_2 \cup V_3$ is a subspace implies that $u+y \in V_1 \cup V_2 \cup V_3$ and $u + 2y \in V_1 \cup V_2 \cup V_3$. If $u + y$ was in V_1, then $y = (u + y) - u$ would be in V_1 which is a contradiction. If $u + 2y$ was in V_1, then $2y = (u+2y)-u$ would be in V_1 which is a contradiction. If $u+y$ was in V_2, then $u = (u + y) - y$ would be in V_2 which is a contradiction. If $u + 2y$ was in V_2, then $u = (u + 2y) - 2y$ would be in V_2 which is a contradiction. Therefore $u + y \in V_3$ and $u + 2y \in V_3$. And it follows that $y = (u + 2y) - (u + y)$ is in V_3.

(iii) We claim that $V_1 \cap V_2 \subset V_3$. Pick $z \in V_1 \cap V_2$. The premise that V_1 is a subspace implies that $u + z \in V_1$. If $u + z \in V_1 \cap V_2$, then $u = (u+z) - z \in V_2$ which is a contradiction. Thus $u+z \in V_1 \setminus (V_1 \cap V_2)$. By (i) we have $u + z \in V_3$. Also by (i) we have $u \in V_3$. The premise that V_3 is a subspace implies that $z = (u + z) - u$ is in V_3.

The above (i) and (iii) implies that $V_1 \subset V_3$. The above (ii) and (iii) implies that $V_2 \subset V_3$.

□

1.36. Exercise 1.C.14

Verify the assertion in Example 1.38: Suppose that $U = \{(x, x, y, y) \in \mathbb{F}^4 : x, y \in \mathbb{F}\}$ and W=$\{(x, x, x, y) \in \mathbb{F}^4 : x, y \in \mathbb{F}\}$. Then

$$U + W = \{(x, x, y, z) \in \mathbb{F}^4 : x, y, z \in \mathbb{F}\}.$$

Proof. (1) Pick $x \in U + W$. Then there exists $u = (x_u, x_u, y_u, y_u) \in U$ and $w = (x_w, x_w, x_w, y_w) \in W$ such that $x_u, y_u, x_w, y_w \in \mathbb{F}$ and $x = u + w$. Thus $x = (x_u + x_w, x_u + x_w, y_u + x_w, y_u + y_w)$, which shows that $x \in \{(x, x, y, z) \in \mathbb{F}^4 : x, y, z \in \mathbb{F}\}$. Therefore $U + W \subset \{(x, x, y, z) \in \mathbb{F}^4 : x, y, z \in \mathbb{F}\}$.

(2) Pick $v \in \{(x, x, y, z) \in \mathbb{F}^4 : x, y, z \in \mathbb{F}\}$. Then there exists $x, y, z \in \mathbb{F}$ such that $v = (x, x, y, z)$. By choosing $u = (x, x, y, y) \in U$ and $w = (0, 0, 0, z - y) \in W$, we have $u + w = v$. Thus $v \in U + W$. Therefore $\{(x, x, y, z) \in \mathbb{F}^4 : x, y, z \in \mathbb{F}\} \subset U + W$.

The above (1) and (2) imply that $U + W = \{(x, x, y, z) \in \mathbb{F}^4 : x, y, z \in \mathbb{F}\}$. □

1.37. Exercise 1.C.15

Suppose U is a subspace of V. What is $U + U$?

PROOF. We claim that $U + U = U$.

(1) Pick $z \in U + U$, then there exists $x, y \in U$ such that $z = x + y$. Since U is closed under addition, we have $x + y \in U$, i.e. $z \in U$. Thus $U + U \subset U$.
(2) Pick $u \in U$, then $u = 1u = (\frac{1}{2} + \frac{1}{2})u = \frac{1}{2}u + \frac{1}{2}u$, i.e. $u \in U + U$. Thus $U \subset U + U$.

The above (1) and (2) imply that $U + U = U$. □

1.38. Exercise 1.C.16

Is the operation of addition on the subspaces of V commutative? In other words, if U and W are subspaces of V, is $U + W = W + U$?

PROOF. We claim that $U + W = W + U$.

(1) Pick $v \in U + W$, then there exists $u \in U$ and $w \in W$ such that $v = u + w$. Since the addition on V is commutative, $u + w = w + u$, and this shows that $v = w + u$ is in $W + U$. Thus $U + W \subset W + U$.
(2) Pick $v \in W + U$, then there exists $w \in W$ and $u \in U$ such that $v = w + u$. Since the addition on V is commutative, $w + u = u + w$, and this shows that $v = u + w$ is in $U + W$. Thus $W + U \subset U + W$.

The above (1) and (2) imply that $U + W = W + U$. □

1.39. Exercise 1.C.17

Is the operation of addition on the subspaces of V associative? In other words, if U_1, U_2, U_3 are subspaces of V, is

$$(U_1 + U_2) + U_3 = U_1 + (U_2 + U_3)?$$

PROOF. We claim that $(U_1 + U_2) + U_3 = U_1 + (U_2 + U_3)$.

(1) Pick $v \in (U_1 + U_2) + U_3$, then there exists $u_1, u_2, u_3 \in U$ such that $v = (u_1 + u_2) + u_3$. Since the addition on V is associative, $(u_1 + u_2) + u_3 = u_1 + (u_2 + u_3)$, and this shows that $v = u_1 + (u_2 + u_3)$ is in $U_1 + (U_2 + U_3)$. Thus $(U_1 + U_2) + U_3 \subset U_1 + (U_2 + U_3)$.
(2) Pick $v \in U_1 + (U_2 + U_3)$, then there exists $u_1, u_2, u_3 \in U$ such that $v = u_1 + (u_2 + u_3)$. Since the addition on V is associative, $u_1 + (u_2 + u_3) = (u_1 + u_2) + u_3$, and this shows that $v = (u_1 + u_2) + u_3$ is in $(U_1 + U_2) + U_3$. Thus $U_1 + (U_2 + U_3) \subset (U_1 + U_2) + U_3$.

The above (1) and (2) imply that $(U_1 + U_2) + U_3 = U_1 + (U_2 + U_3)$. □